

Vaibhav Balloli

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Research Statement

Deployed AI systems fail when the capabilities are not understood by their user: across long conversations, under shifting user intent and environments, and deployment constraints that have additional regulatory complexities and expectations. My work addresses four parts of that gap:

1. Post-training environments for agents to clarify intent, recover, abstain, or delegate.
2. Contextual evaluations built with domain experts replace proxy metrics with judgments that hold.
3. Science of understanding and improving instruction-following behaviors and capabilities of agents
4. Monitoring these systems at test time.

My research has always aimed towards building and deploying human-AI centered approach with stakeholders in challenging domains like reproductive health, wildlife conservation, and road safety - identifying and researching how AI can solve problems, coordinating multi-stakeholder requirements, publishing relevant papers (NeurIPS, COLING, KDD, FAccT, and IJCAI), and deploying them in the real-world.

Research and Engineering Impact

- **HAMS**: Automated driver's-license testing deployed nationally across India; **440,000+ candidates** assessed, adopted by state transport authorities ([live dashboard](#)).
- **PATHFinder**: Conversational LLM agent being deployed with clinicians and patients to personalize pre-natal care planning at the University of Michigan.
- **Elephant Re-ID**: Deploying novel, SOTA computer vision models in Botswana.
- **SmartCampus**: Co-founded a campus payments platform processing **INR 25 million** at **1,000 active users per minute**.

Technical Skills

- **Languages and frameworks**: Python, C++, JavaScript; PyTorch, JAX, SkyRL, verl.

Education

University of Michigan, Ann Arbor (GPA: 4.0)

2023 - Present (expected Dec 2027)

Ph.D. Candidate and M.S., Computer Science and Engineering.

BITS Pilani, Hyderabad Campus

2016 - 2020

B.E., Electronics and Communication Engineering

Research Experience

Microsoft Research Redmond (AI Frontiers), Research Intern

May 2025 – Aug 2025

Advisors: Hussein Mozannar, Adam Fourney, Gagan Bansal, Saleema Amershi, Eric Horvitz

- Applied multi-turn reinforcement learning (GRPO) to improve how LLM agents clarify intent and recover from errors across extended episodes, and built evaluation harnesses for multi-turn agent performance. Under submission with five Microsoft Research co-authors.

Microsoft Research India, Research Fellow

June 2021 – June 2023

Research Fellow (2022–23); Societal Impact through Cloud and AI Fellow (2021–22). Advisors: Dr. Akshay Nambi, Tanuja Ganu, Dr. Venkat Padmanabhan

- **VeLLM**: Built runtime strategies that select prompt, LLM, and embedding model per query, plus the multilingual evaluation and retrieval components of the platform. Evaluated across **18 languages: 10–15% gains** over pre-trained and **4x** over fine-tuned language-specific baselines, with no additional training (COLING 2025, oral).

- **HAMS**: Visual SLAM, object detection, and trajectory analysis for automated driver's-license testing.
- **Chanakya**: Contextual-bandit framework learning runtime accuracy/latency tradeoffs for streaming perception, beating static and dynamic baselines on server GPUs and edge devices (NeurIPS 2023).

Publications and Working Papers

10. **Certifying Long-Context Interactive Abilities of Tool-calling Agents**
Vaibhav Balloli, Elizabeth Bondi-Kelly
Under submission
9. **ELICITBench: When Should Agents Ask? Benchmarking Hidden-State Elicitation in Personalized Search**
Vaibhav Balloli, Elizabeth Bondi-Kelly
Under submission
8. **MATTER-LC: Multimodal Agentic Task Orchestration for Transforming Evidence into Source-Auditable Records in Landslide-Loss Curation**
Xin Wei, Sabine Loos, **Vaibhav Balloli**, Lauren K. Palermo, Benjamin B Mirus, Nathan Wood, Aleeza Wilkins, Elizabeth Bondi-Kelly
7. **RELIANCE: Curating and Evaluating Reproductive Health Information on Social Media**
Vaibhav Balloli, Laura Ellis, Vishala Mishra, Alice Chi, Alex Friedman Peahl, Elizabeth Bondi-Kelly
KDD Datasets and Benchmarks, 2026 · [Project](#)
6. **PATHFinder Agent for Tailored Prenatal Care**
Vaibhav Balloli, Carissa Samuel, Samia Abdelnabi, Alex Peahl, Elizabeth Bondi-Kelly
Interactive Health, 2026 · [Project](#)
5. **“Where is this coming from?” Uncovering Trustworthiness Ideals in AI-powered Peripartum Information Seeking**
Vaibhav Balloli, Julia Erickson, Xinyi Li, Erin van Liemt, Alex Friedman Peahl, Elizabeth Bondi-Kelly
ACM FAccT, 2026
4. **SEEK-CBM: Editable and Interpretable Retrieval for Elephant Re-Identification**
Antoine Salaün, **Vaibhav Balloli**, Timm Haucke, Elizabeth Bondi-Kelly, Sara Beery
CV4Animals Workshop @ CVPR 2026 · [Talk](#)
3. **Bridging the Gap: Dynamic Learning Strategies for Improving Multilingual Performance in LLMs**
Somnath Kumar*, **Vaibhav Balloli***, Mercy Ranjit, Kabir Ahuja, Tanuja Ganu, Sunayana Sitaram, Kalika Bali, Akshay Nambi
COLING 2025 (Oral)
2. **Are They the Same Picture? Adapting Concept Bottleneck Models for Human-AI Collaboration in Image Retrieval**
Vaibhav Balloli, Sara Beery, Elizabeth Bondi-Kelly
IJCAI 2024, Human-Centered AI Track (<5% acceptance) · [Code](#) · [Project](#)
1. **Chanakya: Learning Runtime Decisions for Adaptive Real-Time Perception**
Anurag Ghosh, **Vaibhav Balloli**, Akshay Nambi, Aditya Singh, Tanuja Ganu
NeurIPS 2023 · [Talk](#) · [Code](#)

Awards and Service

- Exceptional rating, Ph.D. Preliminary Exam (advancement to candidacy) 2025
- **2nd place**, Google x MHacks Hackathon (USD 1,500); [featured by labs.google](#) 2024
- Selected for [Harvard/MIT HAIST/MAIA Intro Fellowship](#) on AI Safety 2023
- **Press**: VeLLM cited by [Satya Nadella](#) and in [Times of India](#); HAMS in [Punjab News Express](#) 2022–2024
- **Reviewer**: NeurIPS (2024–2026), KDD 2026, IJCAI 2025, ICML 2025, AAMAS 2025, ICLR 2025; workshops at CVPR 2024, NeurIPS 2023, ACL 2026